Agent

attributes:
- Identification number
- Current Location
- availability
- Health Status

Crew:

attributes:
- Name
- Skills
- Fatigue Level

- Scientist — Scientific-Research
- Egineer — Equipment-repair
- medical officer — medical-care

Robots:

attributes:
- Battery Level

- Exploration Rover — surface-Exploration
- Repair Robot — Equipment-repair
- Drone — Aerial-Recon

facilities:

- name
- Location
- capacity
- operational
- resourses
- equipments

- Habitats
- Laboratories
- Storage facilities
- Power Station
- Communication Station

Resources & Enums

oxygen, water, food, Fuel, Battery power, medical supplies, Spare parts

must be tracked, no mission consume more than available

Storing → Facilities

Rules:

1. Colony wide total
   Total quantities across Colony (check Facillities operational)

2. multiple sources
   mission or operation may draw resource From more than one facility

3. All or nothing Consumption
   if no enough resources → nothing chagnges

4. Local emergencies
   Leak affects the affected habitat only and Total

5. Colony status report
   display total resources

6. Resource allocation policy
   Storage ⟶ habitats

لو ال storage ظعن
habitats الطل و اسحب من ال
Total و ده في حاله ان ال

هيجاني

missions

attributes:

missionState state enums
int duration
int elapsed } unit time each mission duration=3 unit time

List<Agent> assignedAgents = new ArrayList<>();

String failureReason

Map<ResourseType, Integer> required = new HashMap<>();

- **Exploration**
  attributes:
  region
  equipment
  minBattery
  Blocked and failed by dust storm
- **Repair**
  attributes:
  targetFacility or equipment
  repair costs
  fails if target are destroyed
- **Research**
  attributes:
  Laboratory experiment equipment
  Needs operational Lab
- **Rescue**
  attributes:
  Stranded agent
  Needs communication

Rules:

Capability Requirment:

يشيك على instant of skill بدل ما اشوف اي agent اعمل check

one Active mission at a Time:

- only one mission at a time
- mission is submitted to colony with Agents

accepted

state → Assigned and remains active until it completes or fails

Failed conditions

- another mission is active
- agent Lacks the required capability
- required resources are insufficient

Clear reason لازم يكون rejected اي mission ⊗

لو في Fail mission متسجلها ثم خليها في history ⊗

- agents Becomes Free after mission is Finished
- emergency never wait for active mission they work in parallel

Mission ownership

- mission submitted to colony, colony holds → active mission, mission history

- only one mission active at a time
- Agent availability is available();
- mission target or Location → facilities (Repair mission / Research mission), region (Exploration)
- mission resource requirments are checked against the colony wide totals

Mission Execution

- Time start counting when it becomes in progress
- each Time unit advanced mission by 1 unit
- when required Time is reached, mission becomes completed
- when is finished agent becomes avaible

~~for~~ Fails
- insufficient resources
- Equipment failure
- unsuitable agent
- Enviromental condition
- insufficient battery
- unavailable personnal

Mars Enviroment

Enviroment affect Colony operation
- Normal condition
- Dust storm (surface exploration mission can't begin / Active Rover mission fails)
- Communication disruption (drones can't be controlled / certain missions can't be initiated / communication dependent operation may fail)

Emergencies:

Attributes:

id
Location
state
unResolvedReason
responders

- **OxygenLeak**
  - **Attributes:**
    - habitate Habitat
    - LeakAmount int
- **Fire**
  - **Attributes:**
    - Lab Laboratory
    - extraOxygenUse int
- **RoverBreakdown:**
  - **Attributes:**
    - rover (Broken)
    - affectedMission
      - ↓
      - mission it broke during
    - Map<ResourceType, Int> cost = new EnumMap<>
- **medical Emergency:**
  - **Attributes:**
    - crewmember patient
    - int healthrestored
    - facilit treatmentfacility

Locations:

Could be facility or martian surface

agent → Location

rover → Location
- if Breakdown → remains at it place

Location doesn't restrict who can respond

- check responder capability
- check not in mission
- check Can perform mission

Rules:

- each emergency require specific responder
- Emergency dont interrupt mission
- unavailable → unResolved + clear Reason

Oxygen Leaks occures in habitat

- Identify affected habitat
- Reduce oxygen in habitat
- Determine appropriate responders
- Dispatch available responders
- consume required resourses
- repair Leak if possible
- update state of habitat

Oxygen Cost:

Leak detected → oxygen is deducted from habitat
- → Repair → Resolved 1 unit
- → not Repaired → unResolved → deduct again → unResolved 2 units

- oxygen ≥ 0

---

fires (Low , high)

↓ Low status ↓ equipments + need medical assistant

- Laboratory "unavailable"
- Equipment "damaged"
- oxygen consumption "increases"

Requirments → Fire-response personnel
→ medical assistant 1 is enough

Rover Breakdowns

- happens during exploration mission
  - → update rover status
    - → mission status
    - → Rover Location
    - → Required repair resources

---

medical emergency

affected → Crew member

Requirments → medical officer
→ medical supplies
→ habitat should be 'operational' if not then reallocation first

after treatment health ↑

Simulation and Time managment

Class(Simulation

attributes: so simulation points to colony

Colony colony (final)

inT CurrentTime = 0 start

method:

```
advanceTime( int unittime) {
    For( inti=0; i<unittime;i++){
        tick();
    }
}

tick(){
    currenttime+= 1;

print currenttime ---> --- Time 1 ---
                       --- Time 2 ---
```

Colony → updatemission
→ update enviroment
→ updateAgent
→ update Resources

}